Steven Adams

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Education

Sep 2017 - Jun 2021 **MSc. Systems and Control, TU Delft**
- Thesis: Decentralised multi-agent (swarm) foraging with local communication only, published in Autonomous Robots 2023 (Link)
- GPA: 8.2/10

Sep 2018 - Apr 2020 **MSc. Econometrics and Operations Research, Vrije Universiteit Amsterdam**
- Specialisation: Financial Econometrics
- Thesis: Forecasting Cryptocurrency Prices by Non-Gaussian State Space Models with Markov-Switching (Link)
- GPA: 8.4/10

Sep 2013 - Feb 2017 *Active: 3 yr.* **BSc. Mechanical Engineering, TU Delft**
- Minor: Finance (Faculty of Applied Mathematics).
- Honours programme: 9 ECTS including interfaculty project.
- GPA: 8.2/10 (Cum laude)

Work experience

Jun 2026 - present **Freelance Agentic AI Engineer**, *Berlin / Rotterdam*
- Advise and set up agentic solutions for research engineering projects at startups.
- Projects in physics-based modelling, signal processing and calibration of sensing hardware for materials and structures.

Sep 2021 - Jun 2026 **Doctoral Researcher, Delft Center for Systems and Control, TU Delft**
- Member of the TU Delft AI Labs programme (HERALD lab), 5-year funded position.
- Representative on the PhD Council of the Mechanical Engineering faculty.

Nov 2024 - Jan 2025 **Visiting Researcher, University of Oxford**
- Visited the group of Prof. Marta Kwiatkowska, focusing on formal methods for safe AI.

Dec 2019 - Mar 2021 *20 hrs/week* **Trader Assistant - IMC Trading**, *Amsterdam*
- Developed and maintained advanced Python-based reports and dashboards to support real-time trading decisions and strategy optimisation.
- Collaborated closely with traders and analysts to tailor tools to operational needs.

PhD Research

Thesis	Formal Analysis of Stochastic Neural Dynamical Models
Topic	Robust machine learning and uncertainty quantification for safe decision-making
Keywords	Probabilistic & Bayesian ML, Safe AI, Control Theory, Formal Methods

Description In my PhD research, I developed Bayesian and distributional methods to quantify and propagate uncertainty in AI-based control systems. Adopting a formal methods perspective, I derived rigorous guarantees using tools such as optimal transport metrics. My work has potential applications in safety-critical systems such as autonomous driving and robotics.

Publications ICML, JMLR, L-CSS. See Google Scholar (Link).

Awards and Honours

Sep 2024 **Outstanding Teaching Assistant Award**
- Awarded by the Delft Center for Systems and Control, TU Delft.
- Recognised for three years of teaching the MSc course on Control Theory.

Jul 2023 **Outstanding Paper**
- Workshop on Formal Verification of Machine Learning, ICML, 2023.
- Title: Formal Control Synthesis for Stochastic Neural Network Dynamical Models.

Sep 2017 - Sep 2019 *3 hrs/week* **Technology Scholarship - ASML**
- Participation in a personal development programme with bimonthly trainings on both hard and soft skills and mentoring at ASML.
- Receiving financial support.

Interests & Skills

Languages Dutch (native), English (C2), German (B1)

Programming Python, Julia, ROS

Agentic AI Building and shipping with coding agents (Claude Code, OpenCode) and agent SDKs and frameworks (LangGraph), including plugins, skills and subagents; model-agnostic stacks (OpenRouter)